

Advanced Microeconomics

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Problem Set 6: Application of income and substitution effects and welfare measures

1 Application

1) A consumer's utility function is $u = x_1^{1/2} x_2^{1/2}$. She has an income y and prices are p_1 and p_2 . Explore the relationship between utility maximization and expenditure minimization while filling out the values for the given utility function in the table.

Utility Maximizing	Expenditure Minimizing
Marshallian demands	Hicksian demands

Plug in the respective demand functions to get the

indirect utility function	expenditure function
Substitute the expenditure function (e) into the Marshallian demand function to derive the Hicksian demand function	Substitute the indirect utility function (v) into the Hicksian demand function to derive the Marshallian demand function
Invert v and replace I by u to get the expenditure function	Invert e and replace u by v to get the indirect utility function
Check Roy's identity	Check Shephard's lemma

Establish the Slutsky equation

- 2) Suppose there are two goods, prices are given, and there is a consumer with a certain positive income.
- Show in a diagram how it is possible to split the effect of a price decrease on good 1 into the income and substitutions effects. Assume that the good is a normal good.
 - If the good had been an inferior good, what would have been different in the graph.
 - If the good had been an Giffen good, what would have been different.
- 3) [OPTIONAL] A consumer has preferences over the single good x and all other goods m represented by the utility function, $u(x, m) = \ln(x) + m$. Let the price of x be p , the price of m be unity, and let income be I .
- Derive the Marshallian demands for x and m .
 - Derive the indirect utility function, $v(p, I)$.
 - Use the Slutsky equation to decompose the effect of an own-price change on the demand for x into an income and substitution effect. Interpret your result briefly.
 - Suppose that the price of x rises from p^0 to $p^1 > p^0$. Show that the consumer surplus area between p^0 and p^1 gives an exact measure of the effect of the price change on consumer welfare.
 - Carefully illustrate your findings with a set of two diagrams: one giving the indifference curves and budget constraints on top, and the other giving the Marshallian and Hicksian demands below. Be certain that your diagrams reflect all qualitative information on preferences and demands that you've uncovered. Be sure to consider the two prices p^0 and p^1 , and identify the Hicksian and Marshallian demands.